

$$b) \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} = \frac{e^0 - 1}{0} = \frac{1-1}{0} = \frac{0}{0} \quad \times$$

$$\frac{d}{dx} = \frac{e^x - 0}{3x^2} = \frac{e^0 - 0}{3(0)} = \frac{1}{0} = \infty \quad \text{Ans}$$

$$a) \quad \lim_{x \rightarrow \infty} \frac{x^{-1/3}}{\sin(\frac{1}{x})} = \frac{(\frac{1}{\infty})}{\sin(0)} = \frac{0}{0} \quad \times \quad \therefore \frac{1}{\infty} = 0$$

$$\frac{d}{dx} = \frac{-\frac{1}{3}x^{-\frac{2}{3}}}{\cos(\frac{1}{x})(-\frac{1}{x^2})} = \frac{(\frac{1}{\infty})}{(1)(\frac{1}{\infty})} = \frac{0}{0} \quad \times$$

$$= \frac{-\frac{1}{3}x^{-\frac{2}{3}}}{\cos(\frac{1}{x})x(\frac{1}{x^2})}$$

$$= \frac{-\frac{1}{3}x^{-\frac{2}{3}} \times x^2}{\cos(\frac{1}{x})}$$

$$= \frac{-\frac{1}{3}x^{\frac{1}{3}}}{\cos(\frac{1}{x})}$$

$$= \frac{-\frac{1}{3}x^{\frac{1}{3}}}{\cos(\frac{1}{\infty})}$$

$$= \frac{x^{-\frac{1}{3}}}{\cos(\frac{1}{\infty})}$$

$$= \frac{(\frac{1}{\infty})}{\cos(0)}$$

$$= \frac{0}{1}$$

$$= 0 \quad \text{Ans}$$

$$\frac{0}{0} \text{ indeterminate, } \frac{\infty}{\infty}$$

$$\frac{1}{0} = \infty, \quad \frac{0}{1} = 0$$

$$e^x = x \quad x^{x>1} = +\infty$$

$$e^{-x} = 0 \quad ? \quad x^{x<1} = 0$$

$$b) \quad \lim_{x \rightarrow 0} \frac{\ln x}{\operatorname{cosec} x}$$

$$2) \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x} = \frac{1-1}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{e^x - (-1)e^{-x}}{\cos x}$$

$$= \frac{e^x + e^x}{\cos x} = \frac{1+1}{1} = 2$$

$$2) \lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{\cos x - 1} = \frac{e^0 - 1}{1-1} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{e^{x^2} \times 2x - 0}{-\sin x - 0}$$

$$= \frac{2x e^{x^2}}{-\sin x} = \frac{0}{0} \times$$

$$\frac{d^2y}{dx^2} = \frac{2(x \cdot e^{x^2})}{-\sin x}$$

$$= \frac{2(x \cdot 2x e^{x^2} + e^{x^2} \cdot (1))}{-\cos x}$$

$$= \frac{2(0+1)}{-1}$$

$$= -2$$

$$c) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^2 \tan x} = \frac{0-0}{0} = \frac{0}{0} \times$$

$$= \frac{\tan x - \sin x}{x^2 \cdot \tan x}$$

$$= \frac{\frac{\sin x}{\cos x} - \sin x}{x^2 \cdot \frac{\sin x}{\cos x}}$$

$$= \frac{\frac{\sin x - \sin x \cos x}{\cos x}}{x^2 \cdot \frac{\sin x}{\cos x}}$$

$$= \frac{\sin x - \sin x \cos x}{x^2 \cdot \sin x}$$

$$= \frac{\sin x (1 - \cos x)}{x^2 \sin x}$$

$$= \frac{1 - \cos x}{x^2}$$

$$\frac{dy}{dx} = \frac{0 - (-\sin x)}{2x}$$

$$= \frac{\sin x}{2x} \quad \therefore \frac{\sin x}{x} = 1$$

$$= \frac{1}{2}$$

$$d) \lim_{x \rightarrow 0} \frac{x - \tan x}{x - \sin x} = \frac{0-0}{0-0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{1 - \sec^2 x}{1 - \cos x}$$

$$= 1 - \frac{1}{\cos^2 x} \div 1 - \cos x$$

$$= 1 - \frac{1}{1} \div 1 - 1 = \frac{0}{0} \times$$

$$= \frac{1 - \sec^2 x}{1 - \cos x}$$

$$= \frac{1 - 2(\sec x)(\sec x \cdot \tan x)}{1 - \cos x}$$

$$= \frac{1 - 2\left(\frac{1}{\cos x}\right)\left(\frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}\right)}{1 - \cos x}$$

$$= \frac{1 - 2(1)(0)}{1 - 1} = \frac{1}{0}$$

$$1 - \sec^2 x \div 1 - \cos x$$

$$0 - 2 \sec x \tan x \div 0 - (-\sin x)$$

$$= \frac{-2 \sec^2 x \tan x}{\sin x}$$

$$= - \frac{2 \cdot \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}}{\sin x}$$

$$= \frac{-2 \sin x}{\cos^2 x} \times \frac{1}{\sin x}$$

$$= - \frac{2}{1}$$

$$=$$

$$e) \quad \lim_{x \rightarrow 0} \frac{\ln(\sin 3x)}{\ln(\sin x)}$$

$$\sin 3x$$

$$= \ln(\sin 3x) \div \ln(\sin x)$$

$$3x \cos 3x$$

$$= \frac{1}{\sin 3x} \times 3 \cos 3x \div \frac{1}{\sin x} \times \cos x$$

$$= \frac{3 \cos 3x}{\sin 3x} \div \frac{\cos x}{\sin x}$$

$$= \frac{3}{\tan 3x} \div \frac{1}{\tan x}$$

$$= \frac{3}{\tan 3x} \times \tan x$$

$$= \frac{3 \tan x}{\tan 3x}$$

$$= \frac{3 \sec^2 x}{3 \sec^2 3x}$$

$$= \frac{3}{3}$$

$$= 1$$

$$e) \lim_{x \rightarrow 0} \frac{x - \tan x}{x - \sin x} = \frac{0-0}{0-0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = x - \tan x \div x - \sin x$$

$$= 1 - \sec^2 x \div 1 - \cos x$$

$$= \frac{0}{0} \times$$

$$= \frac{1 - \sec^2 x}{1 - \cos x}$$

$$= \frac{0 - 2 \sec^2 x \cdot \tan x}{0 + (-\sin x)}$$

$$= \frac{2 \sec^2 x \cdot \tan x}{-\sin x}$$

$$= \frac{2(0) \cdot 0}{0} = \frac{0}{0} \times$$

$$= \frac{2 (\sec^2 x \cdot \sec^2 x + \tan x \cdot 2 \sec^2 x \cdot \tan x)}{-\cos x}$$

$$= \frac{2 (\sec^4 x + 2 \sec^2 x \tan^2 x)}{-\cos x}$$

$$= \frac{2 (1 + 0)}{-1}$$

$$= -2$$

SS Notes

$$a) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \frac{1 - 1}{1 - 1} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{2x - 0}{1 - 0} = \frac{2x}{1} = 2(1) = 2$$

$$b) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\sin(0)}{(0)} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{\cos x}{1} = \cos(0) = 1$$

$$c) \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = \frac{\sin(2(0))}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{\cos 2x \times 2}{1} = \cos(0) \times 2 = 2$$

$$d) \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\cos x} = \frac{1 - 1}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{0 - \cos x}{-\sin x} = \frac{0}{1} = 0$$

$$e) \lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} = \frac{e^0 - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{e^x - 0}{3x^2} = \frac{e^0}{3(0)} = \frac{1}{0} = \infty$$

$$a) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1 - 1}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{0 - (-\sin x)}{2x} = \frac{\sin(0)}{2(0)} = \frac{0}{0} \times$$

$$\frac{d^2 y}{dx^2} = \frac{\cos x}{2} = \frac{1}{2}$$

$$a) \lim_{x \rightarrow 0} \frac{\tan x}{x^2} = \frac{\tan(0)}{0} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{\sec^2 x}{2x} = \frac{1}{\frac{(\cos(0))^2}{0}} = \frac{1}{0} = \infty$$

$$a) \lim_{x \rightarrow \infty} \frac{x^{-\frac{1}{3}}}{\sin(\frac{1}{x})} = \frac{\infty^{-\frac{1}{3}}}{\sin(\frac{1}{\infty})} = \frac{0}{\sin(0)} = \frac{0}{0} \times$$

$$\frac{dy}{dx} = \frac{-\frac{1}{3}x^{-\frac{2}{3}}}{\cos(\frac{1}{x})} \cdot \frac{1}{x} = \frac{-\frac{1}{3}x^{-\frac{2}{3}}}{\cos(\frac{1}{x})} \cdot x$$

$$= \frac{-\frac{1}{3}x^{-\frac{2}{3}}}{\cos(\frac{1}{x})} = \frac{x^{-\frac{2}{3}}}{\cos(0)} = \frac{0}{1} = 0$$

$$a) \lim_{x \rightarrow 0} \frac{\ln x}{\cos x} = \frac{\ln(0)}{\frac{1}{\sin(0)}} = \frac{\ln(0)}{\frac{1}{0}} = \frac{\ln(0)}{\infty} \times$$

$$\frac{dy}{dx} = \frac{\frac{1}{x}}{-\cot x \cdot \cos x}$$

$$= \frac{1}{x} \div -\frac{1}{\tan x} \cdot \frac{1}{\sin x}$$

$$= \frac{1}{x} \div -\frac{\cos x}{\sin x \cdot \sin x}$$

$$= \frac{\sin x \cdot \sin x}{\cos x \cdot x} \quad \therefore \frac{\sin x}{x} = 1$$

$$= \frac{\sin(0)}{\cos(0)} \cdot 1$$

$$= \frac{0}{1} \cdot 1 = 0 \times 1 = 0$$

$\frac{0}{0}$ indeterminate, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty \cdot 0$

$$\frac{1}{0} = \infty, \frac{0}{1} = 0$$

$$e^x = \infty \quad x > 0 \rightarrow +\infty$$

$$e^{-x} = 0 \quad ? \quad x < 0 \rightarrow 0$$

$$\frac{1}{\infty} = 0$$

$$b) \lim_{x \rightarrow 0} \frac{x \cdot \ln x}{x} = 0 \cdot \infty \quad \times$$

$$= \frac{\ln x}{\frac{1}{x}}$$

$$(u'v + uv')$$

$$\frac{dy}{dx} = \frac{1}{x} \div \frac{1}{x^2}$$

$$0 \cdot \frac{1}{x} + \ln x \cdot 1$$

$$= \frac{x^2}{x}$$

$$\frac{dy}{dx}$$

$$1 + \ln x$$

$$= 0$$

$$\frac{d^2y}{dx^2}$$

$$= 0 + \frac{1}{x}$$

$$= \frac{1}{0} = \infty$$

$$\infty^0, 0^0, 1^\infty, \frac{0}{0} \text{ indeterminate}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty \cdot 0$$

$$\frac{1}{0} = \infty, \frac{0}{\infty} = 0$$

$$c) \lim_{x \rightarrow \frac{\pi}{4}} (1 - \tan x) \sec 2x = (1 - 1) \cdot \left(\frac{1}{0} \right)$$

$$= 0 \cdot \infty \quad \times$$

$$e^x = x$$

$$x^{x^0} = +\infty$$

$$e^{-x} = 0$$

$$x^{x^0} = 0$$

$$\frac{1}{0} = 0$$

$$= \frac{1 - \tan x}{\cos 2x}$$

$$\frac{dy}{dx} = \frac{0 - \sec^2 x}{-2 \sin 2x}$$

$$= \frac{1}{\cos^2 x} \div 2 \sin 2x$$

$$= \frac{1}{2} \div 2(1)$$

$$= \frac{2}{2} = 1$$

$$d) \lim_{x \rightarrow 0} \frac{\sin x - x}{x \sin x} = \frac{\sin(0) - 0}{0(\sin 0)} = \frac{0}{0} \quad \times$$

$$\frac{dy}{dx} = \frac{\cos x - 1}{x \cos x + \sin x} = \frac{1 - 1}{0 + 0} = \frac{0}{0} \quad \times$$

$$= \frac{-\sin x}{-x \sin x + \cos x + \cos x}$$

$$= \frac{-\sin x}{-x \sin x + 2 \cos x} = \frac{0}{0 + 2} = \frac{0}{2} = 0$$

$$\textcircled{a}) \quad \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}} = (1+0)^{\infty} = 1^{\infty} \times$$

$$y = (1 + \sin x)^{\frac{1}{x}}$$

$$\ln y = \ln (1 + \sin x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (1 + \sin x)$$

$$= \frac{\ln (1 + \sin x)}{x}$$

$$= \frac{1}{(1 + \sin x)} \times (0 + \cos x) \neq 1$$

$$\ln y = \frac{\cos x}{1 + \sin x}$$

$$\ln y = \frac{1}{1}$$

$$y = e^1$$

$$y = e$$

